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Faculty of Mechanical and Industrial Engineering

Fluid Mechanics (MEng3133)

Chapter TWO Fluid Statics

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Outline

- ❖ Introduction
- ❖ Hydrostatics pressure in Fluids;
- ❖ Pressure measurements;
- ❖ Manometers;
- ❖ Hydrostatics force on plane surfaces;
- ❖ Buoyancy; Flotation and stability;
- ❖ Pressure distribution in rigid body motion.



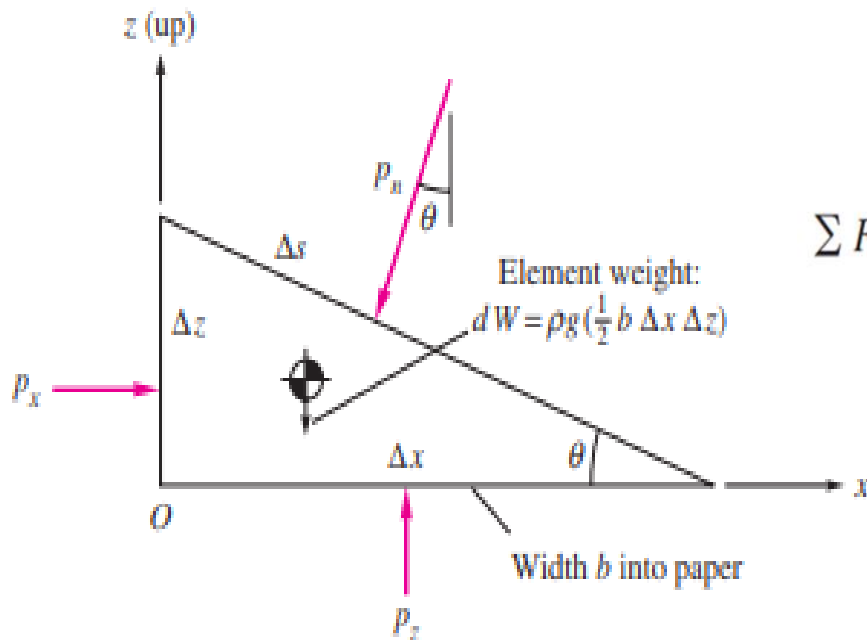
Introduction

- Fluid statics is generally referred to as **hydrostatics** when the fluid is a liquid and as **aerostatics** when the fluid is a gas.
- When the **fluid velocity is zero** and **in rigid-body motion**, denoted as the hydrostatic condition.
- In fluid statics
 - ❑ There is no relative motion between adjacent fluid layers,
 - ❑ There **are no shear (tangential) stresses** in the fluid trying to deform it.
 - ❑ The only stress we deal with in fluid statics is the **normal stress**, which is the pressure, and the variation of pressure is due only to the weight of the fluid.



Hydrostatics pressure in Fluids

- “The intensity of pressure at any point in a liquid at rest, is the same in all direction.” A fluid at rest cannot support shear stress. the normal stress on any plane through a fluid element at rest is a point property called the **fluid pressure**, p .



$$\sum F_x = 0 = p_x b \Delta z - p_n b \Delta s \sin \theta$$

$$\sum F_z = 0 = p_z b \Delta x - p_n b \Delta s \cos \theta - \frac{1}{2} \rho g b \Delta x \Delta z$$

$$\Delta s \sin \theta = \Delta z \quad \Delta s \cos \theta = \Delta x$$

$$p_x = p_n \quad p_z = p_n + \frac{1}{2} \rho g \Delta z$$



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- These relations illustrate two important principles of the hydrostatic, or shear-free, condition:

- (1) There is no pressure change in the horizontal direction, and
 - (2) there is a vertical change in pressure proportional to the density, gravity, and depth change.
- $$\Delta z \rightarrow 0 \quad p_x = p_z = p_n = p$$

Since θ is arbitrary, we conclude that the pressure p in a static fluid is a point property, independent of orientation.

“Any two points at the same elevation in a continuous mass of the same static fluid will be at the same pressure. This is called Pascal’s law.”

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$



Hydrostatic law

- “Pressure in a continuously distributed uniform static fluid varies only with vertical distance and is independent of the shape of the container. The pressure is the same at all points on a given horizontal plane in the fluid. The pressure increases with depth in the fluid.”
- Pressure (or any other stress, for that matter) causes a net force on a fluid element when it varies spatially.

$$p = p(x, y, z, t)$$

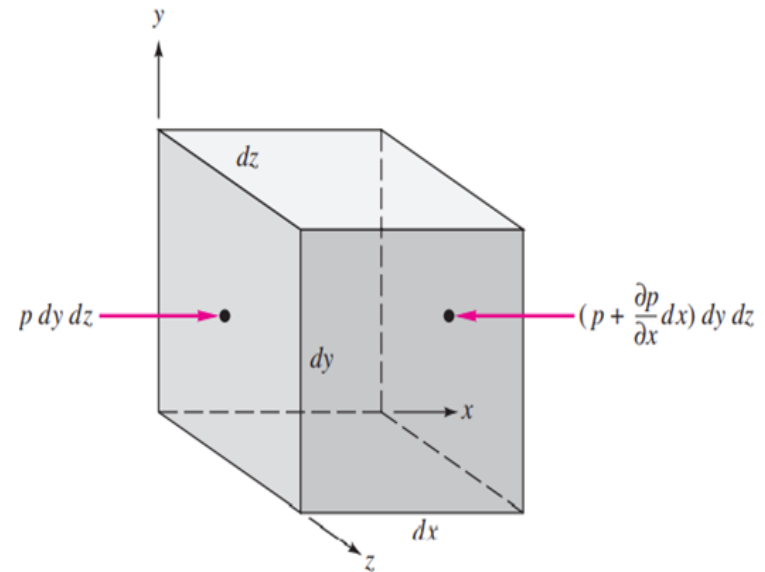
$$dF_x = p \, dy \, dz - \left(p + \frac{\partial p}{\partial x} dx \right) dy \, dz = -\frac{\partial p}{\partial x} dx \, dy \, dz$$



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- $P = P(x, y, z, t)$
- $dF_x = p dy dz - \left(p + \frac{\partial p}{\partial x} dx \right) dy dz = -\frac{\partial p}{\partial x} dx dy dz$
- $dF_y = -\frac{\partial p}{\partial y} dx dy dz, dF_z = -\frac{\partial p}{\partial z} dx dy dz$
- The net force per unit volume is

$$\mathbf{f}_{press} = -\nabla p$$





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- Thus it is not the pressure but the pressure gradient causing a net force that must be balanced by gravity or acceleration or some other effect in the fluid.
- The forces act on the fluid element are;
 - ❖ Surface force
 - Pressure force
 - Viscous force
 - ❖ Body force
 - Gravitational force(weight of element)

$$d\mathbf{F}_{\text{grav}} = \rho \mathbf{g} \, dx \, dy \, dz$$

$$\mathbf{f}_{\text{grav}} = \rho \mathbf{g}$$



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- By Newton's law the sum of these per-unit-volume forces equals the mass per unit volume (density) times the acceleration a of the fluid element:

$$\sum \mathbf{f} = \mathbf{f}_{\text{press}} + \mathbf{f}_{\text{grav}} + \mathbf{f}_{\text{visc}} = -\nabla p + \rho \mathbf{g} + \mathbf{f}_{\text{visc}} = \rho \mathbf{a}$$

- **Hydrostatics, the viscous stresses and the acceleration are zero.** The equation for the pressure distribution reduces to

$$\nabla p = \rho \mathbf{g}$$

- $\frac{\partial p}{\partial x} = 0, \frac{\partial p}{\partial y} = 0, \frac{\partial p}{\partial z} = -\rho g = -\gamma,$
- $\frac{dp}{dz} = -\rho g, p_2 - p_1 = -\int_1^2 \gamma dz,$

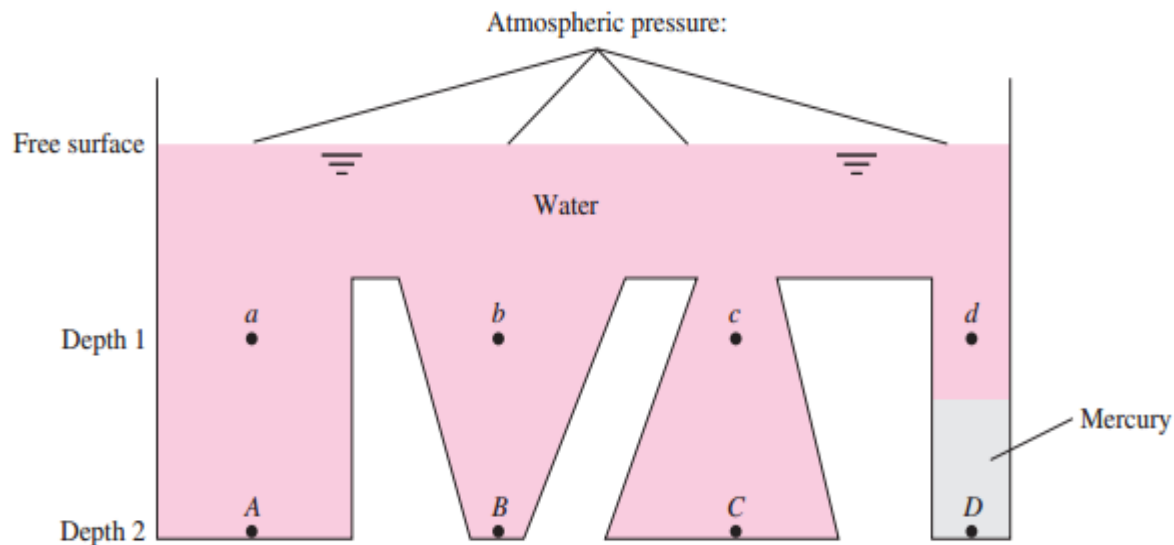
For Gases

$$\begin{aligned} \frac{dp}{dz} &= -\rho g = -\frac{p}{RT} g \\ \int_1^2 \frac{dp}{p} &= \ln \frac{p_2}{p_1} = \frac{g}{R} \int_1^2 \frac{dz}{T} \\ p_2 &= p_1 e^{\left[\frac{g(z_2 - z_1)}{RT_0} \right]} \end{aligned}$$



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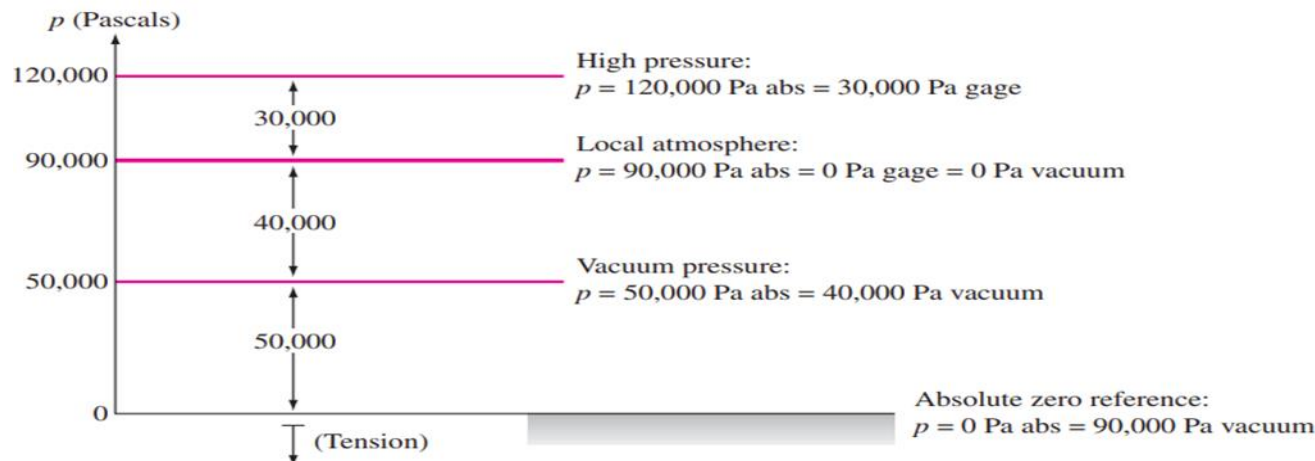
- Liquids are so nearly incompressible that we can neglect their density variation in hydrostatics.
- $p_2 - p_1 = -\gamma(z_2 - z_1), \quad \frac{p_2 - p_1}{\gamma} = (z_1 - z_2), \quad p = p_a + h\gamma$





Pressure Measurements

- ❖ The absolute or total magnitude
- ❖ The value relative to the local ambient atmosphere (**Gage Pressure and Vacuum Pressure**). because many pressure instruments are of differential type and record, not an absolute magnitude, but the difference between the fluid pressure and the atmosphere.
 1. $p > p_a$ *Gage pressure:* $p(\text{gage}) = p - p_a$
 2. $p < p_a$ *Vacuum pressure:* $p(\text{vacuum}) = p_a - p$





The Mercury Barometer

- Atmospheric pressure is measured by a device thus, the atmospheric pressure is often referred to as the barometric pressure called a **barometer**. From the above equation,

$$p_2 - p_1 = -\gamma(z_2 - z_1)$$

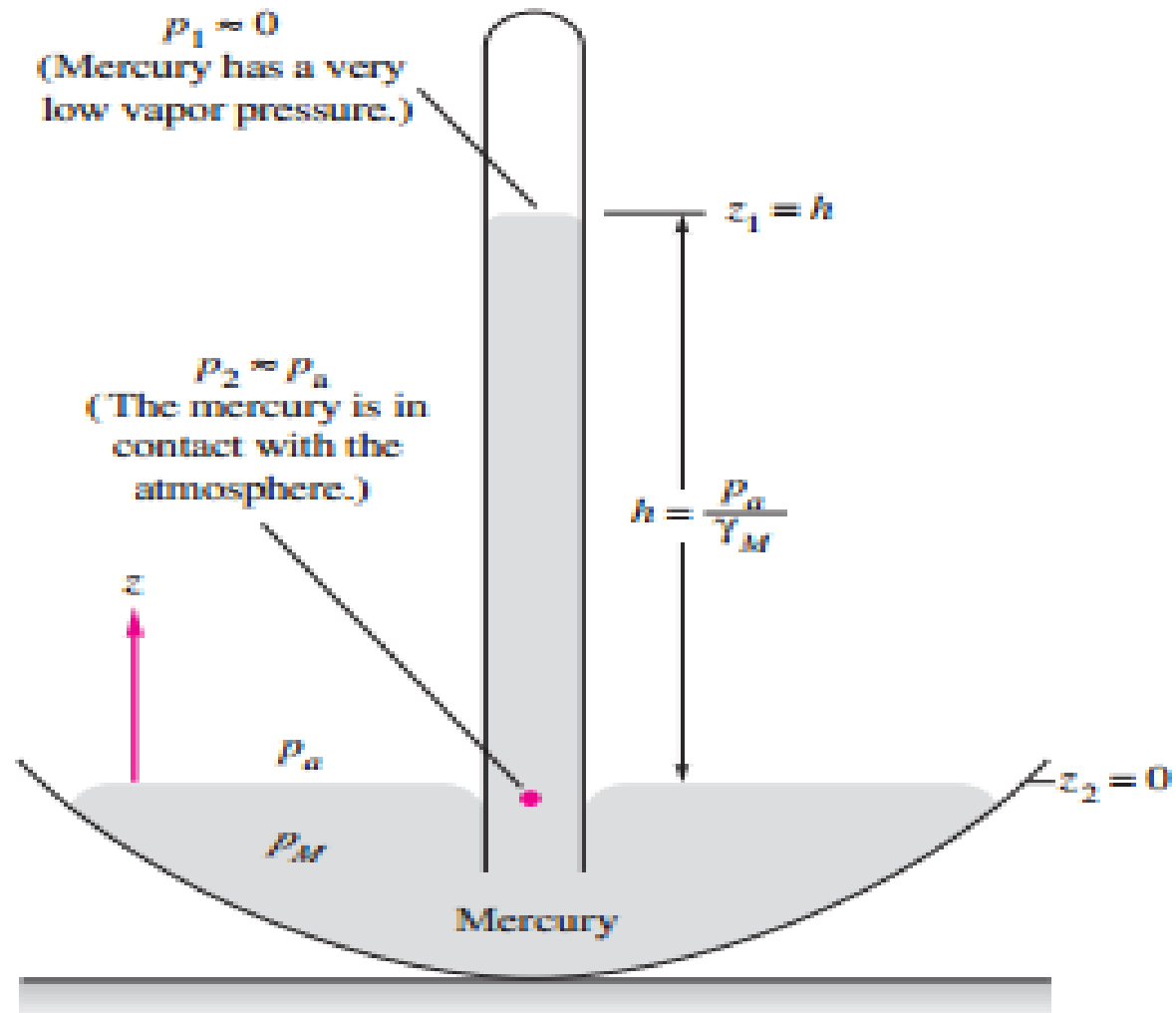
$$p_a - p_1 = -\gamma(0 - h)$$

$$p_a = \gamma_M h$$

- At sea-level standard, with $p_a = 101,350$ Pa and $\gamma_M = 133,100$ N/m³ the barometric height is $h = 101,350 / 133,100 = 0.761$ m or 761 mm.



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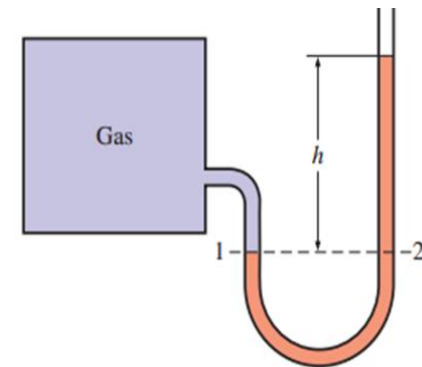




Application to Manometry

- Manometer is a device used to measure pressure differences between two points using static column of one or more liquids or gases.
- Since the gravitational effects of gases are negligible, the pressure anywhere in the tank and at position 1 has the same value see figure below. Furthermore, since pressure **in a fluid does not vary in the horizontal direction within a fluid**, the pressure at point 2 is the same as the pressure at point 1.

$$P_2 = P_{\text{atm}} + \rho gh$$

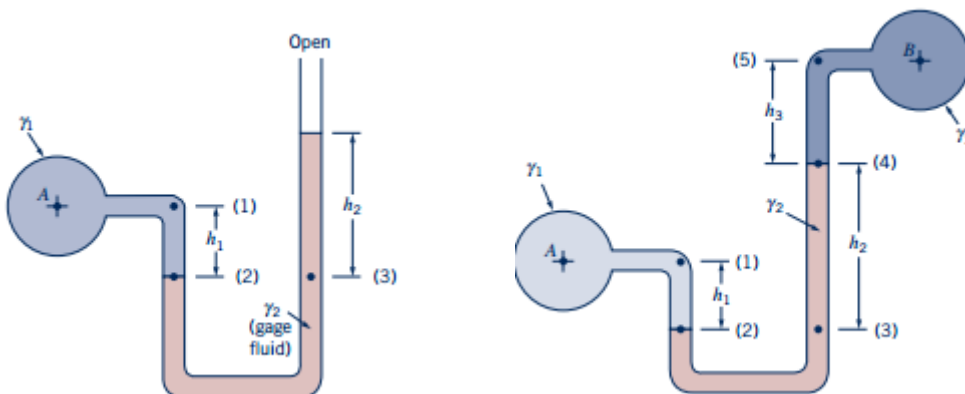




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Manometer Rule

1. Write the pressure at either end of the manometer;
2. Proceed through the manometer, adding γh if moving to a greater depth or subtracting γh if moving to a lesser depth;
3. Stop at the far end, or any point in between, and set the expression equal to the local pressure;

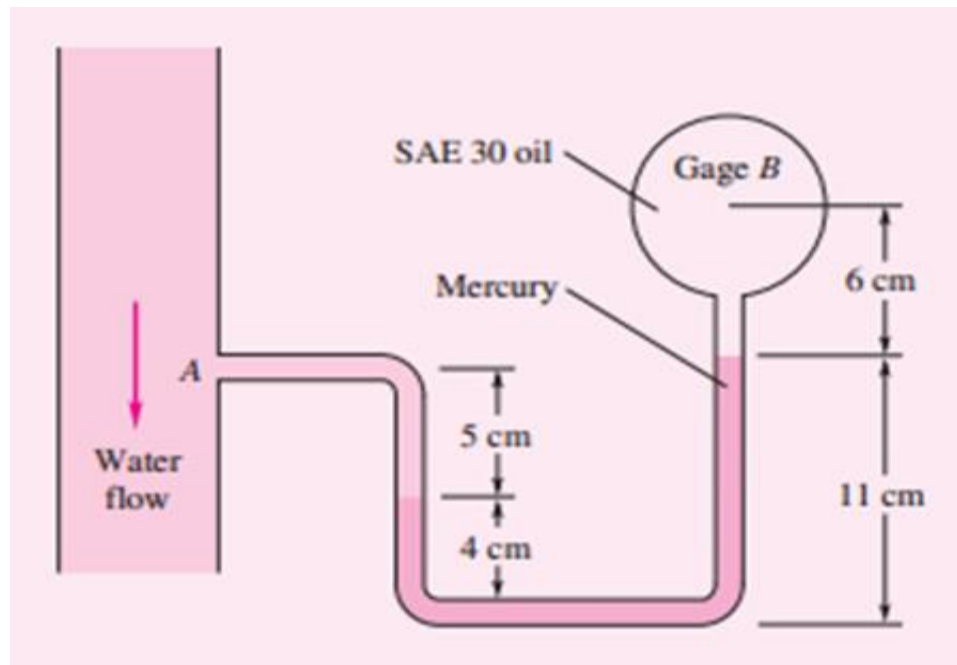




Examples

$$P_A = P_{\text{atm}} - \gamma_1 h_1 + \gamma_2 h_2 \quad P_A - P_B = \gamma_3 h_3 + \gamma_2 h_2 - \gamma_1 h_1.$$

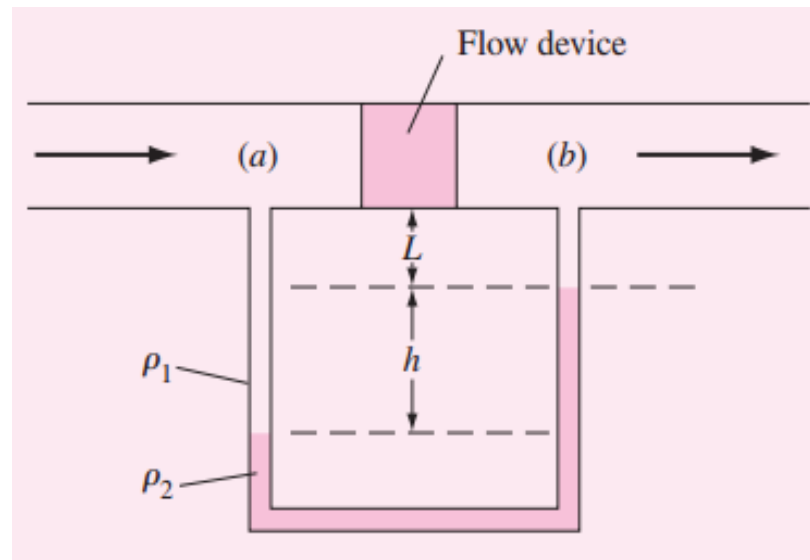
1. Pressure gage B is to measure the pressure at point A in a water flow. If the pressure at B is 87 kPa, estimate the pressure at A in kPa. Assume all fluids are at 20 °C.





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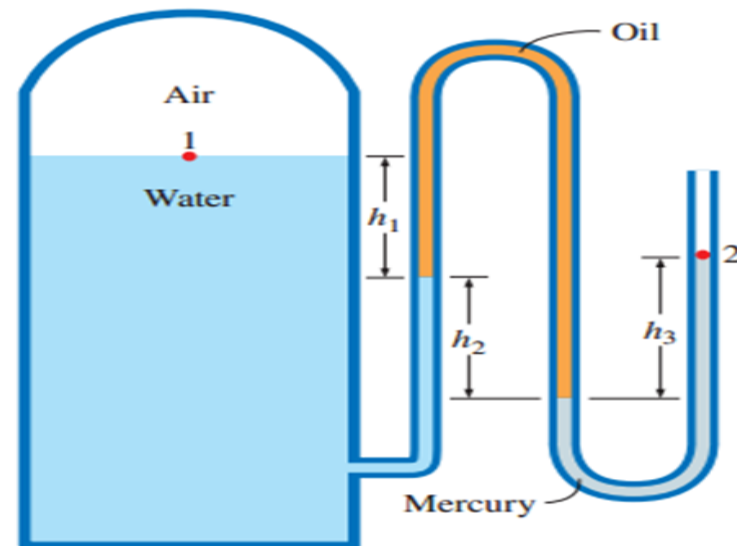
2. The classic use of a manometer is when two U-tube legs are of equal length, as shown in figure below, and the measurement involves a pressure difference across two horizontal points. The typical application is to measure pressure change across a flow device, as shown. Derive a formula for the pressure difference $p_a - p_b$ in terms of the system parameters.





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3. The water in a tank is pressurized by air, and the pressure is measured by a multi fluid manometer as shown in Figure below. The tank is located on a mountain at an altitude of 1400 m where the atmospheric pressure is 85.6 kPa. Determine the air pressure in the tank if $h_1 = 0.1$ m, $h_2 = 0.2$ m, and $h_3 = 0.35$ m. Take the densities of water, oil, and mercury to be 1000 kg/m^3 , 850 kg/m^3 , and $13,600 \text{ kg/m}^3$, respectively.





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• Solution to examples

$$\gamma_{\text{water}} = 9790 \text{ N/m}^3; \quad \gamma_{\text{mercury}} = 133,100 \text{ N/m}^3; \quad \gamma_{\text{oil}} = 8720 \text{ N/m}^3$$

- *Solution steps:* Proceed from A to B, “down” then “up,” jumping across at the left mercury meniscus:

$$p_A + \gamma_w |\Delta z|_w - \gamma_m |\Delta z|_m - \gamma_o |\Delta z|_o = p_B$$

$$\text{or } p_A + (9790 \text{ N/m}^3)(0.05 \text{ m}) - (133,100 \text{ N/m}^3)(0.07 \text{ m}) - (8720 \text{ N/m}^3)(0.06 \text{ m}) = 87,000$$

$$\text{or } p_A + 490 - 9317 - 523 = 87,000 \text{ Solve for } p_A = 96,350 \text{ N/m}^2 \approx 96.4 \text{ kPa} \quad \text{Ans.}$$

$$P_1 + \rho_{\text{water}}gh_1 + \rho_{\text{oil}}gh_2 - \rho_{\text{mercury}}gh_3 = P_2 = P_{\text{atm}}$$

Solving for P_1 and substituting,

$$\begin{aligned} P_1 &= P_{\text{atm}} - \rho_{\text{water}}gh_1 - \rho_{\text{oil}}gh_2 + \rho_{\text{mercury}}gh_3 \\ &= P_{\text{atm}} + g(\rho_{\text{mercury}}h_3 - \rho_{\text{water}}h_1 - \rho_{\text{oil}}h_2) \\ &= 85.6 \text{ kPa} + (9.81 \text{ m/s}^2)[(13,600 \text{ kg/m}^3)(0.35 \text{ m}) - (1000 \text{ kg/m}^3)(0.1 \text{ m}) \\ &\quad - (850 \text{ kg/m}^3)(0.2 \text{ m})]\left(\frac{1 \text{ N}}{1 \text{ kg}\cdot\text{m/s}^2}\right)\left(\frac{1 \text{ kPa}}{1000 \text{ N/m}^2}\right) \\ &= \mathbf{130 \text{ kPa}} \end{aligned}$$

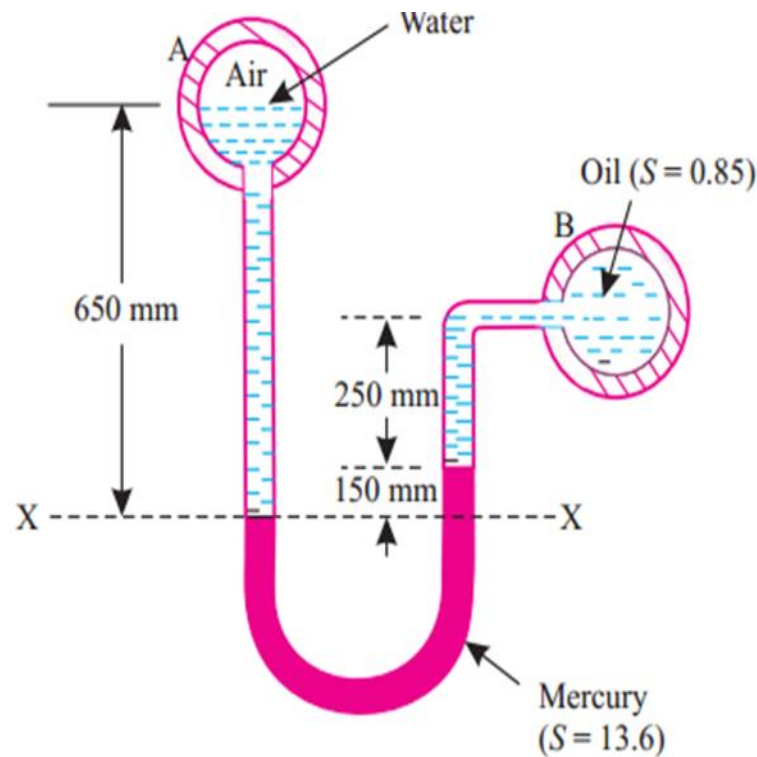
$$p_a + \rho_1gL + \rho_1gh - \rho_2gh - \rho_1gL = p_b$$

$$p_a - p_b = (\rho_2 - \rho_1)gh$$



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Exercise 1. Figure below shows a differential manometer connected at two points A and B. At A air pressure is 100 kN/m^2 . Find the absolute pressure at point B. (**answer, $P_B = 84.36 \text{ kPa}$**)





Hydrostatic Forces on Plane Surfaces

- The design of containment structures requires computation of the **hydrostatic forces** on various solid surfaces adjacent to the fluid. These forces relate to the weight of fluid bearing on the surface.
- A plate (such as a gate valve in a dam, the wall of a liquid storage tank, or the hull of a ship at rest) is subjected to fluid pressure distributed over its surface when exposed to a liquid.
- On a plane surface, the hydrostatic forces form a system of parallel forces, and we often need to determine the **magnitude of the force** and its point of application, which is called **the center of pressure**.



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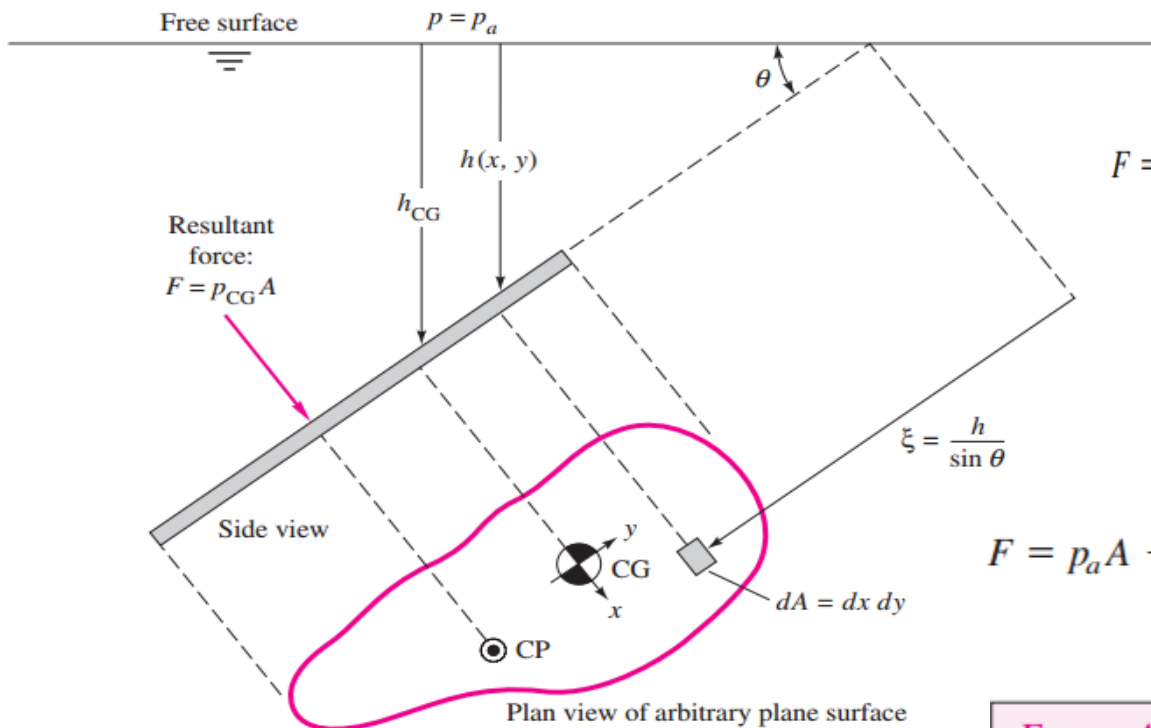
- **Total pressure;** is defined as the force exerted by a static fluid on a surface (either plane or curved) when the fluid comes in contact with the surface. This force is always at right angle (normal)to the surface.



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Hydrostatic Force on Inclined Surfaces

Consider an arbitrary plane surface submerged in liquid in the figure below;



$$F = \int p dA = \int (p_a + \gamma h) dA = p_a A + \gamma \int h dA$$

$$\xi_{CG} = \frac{1}{A} \int \xi dA$$

$$F = p_a A + \gamma \sin \theta \int \xi dA = p_a A + \gamma \sin \theta \xi_{CG} A$$

$$F = p_a A + \gamma h_{CG} A = (p_a + \gamma h_{CG}) A = p_{CG} A$$



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- However, to balance the bending-moment portion of the stress, the resultant force $\mathbf{F}$ acts not through the centroid but below it toward the high-pressure side. Its line of action passes through the center of pressure CP of the plate.
- To find the coordinates (x_{CP}, y_{CP}) , we sum moments of the elemental force $p dA$ about the centroid and equate to the moment of the resultant F . To compute y_{CP} , we equate

$$F y_{CP} = \int y p dA = \int y (p_a + \gamma \xi \sin \theta) dA = \gamma \sin \theta \int y \xi dA$$

$$\xi = \xi_{CG} - y,$$

- The term $\int p_a y dA$ vanishes

$$F y_{CP} = \gamma \sin \theta \left(\xi_{CG} \int y dA - \int y^2 dA \right) = -\gamma \sin \theta I_{xx}$$



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- The determination of x_{CP} is exactly similar:

$$\begin{aligned} Fx_{CP} &= \int xp \, dA = \int x[p_a + \gamma(\xi_{CG} - y) \sin \theta] \, dA \\ &= -\gamma \sin \theta \int xy \, dA = -\gamma \sin \theta I_{xy} \end{aligned}$$

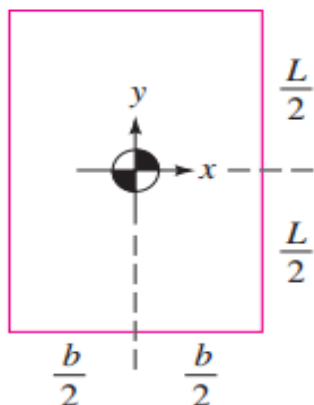
$$y_{CP} = -\gamma \sin \theta \frac{I_{xx}}{\rho_{CG} A}$$

$$x_{CP} = -\gamma \sin \theta \frac{I_{xy}}{\rho_{CG} A}$$

$$F = \gamma h_{CG} A \quad y_{CP} = -\frac{I_{xx} \sin \theta}{h_{CG} A} \quad x_{CP} = -\frac{I_{xy} \sin \theta}{h_{CG} A}$$



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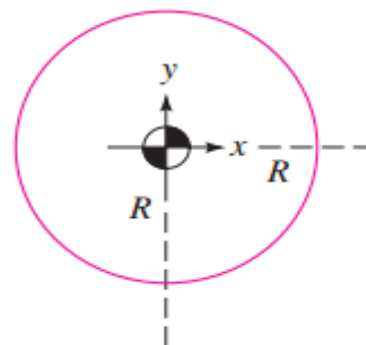


(a)

$$A = bL$$

$$I_{xx} = \frac{bL^3}{12}$$

$$I_{xy} = 0$$

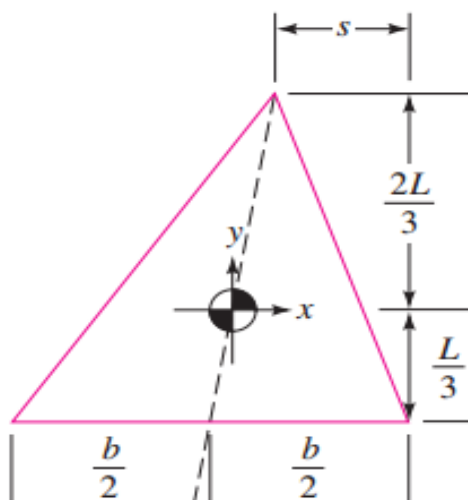


(b)

$$A = \pi R^2$$

$$I_{xx} = \frac{\pi R^4}{4}$$

$$I_{xy} = 0$$

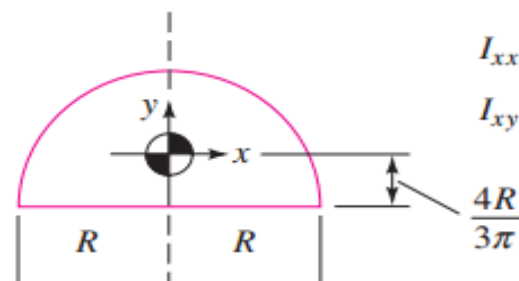


(c)

$$A = \frac{bL}{2}$$

$$I_{xx} = \frac{bL^3}{36}$$

$$I_{xy} = \frac{b(b-2s)L^2}{72}$$



(d)

$$A = \frac{\pi R^2}{2}$$

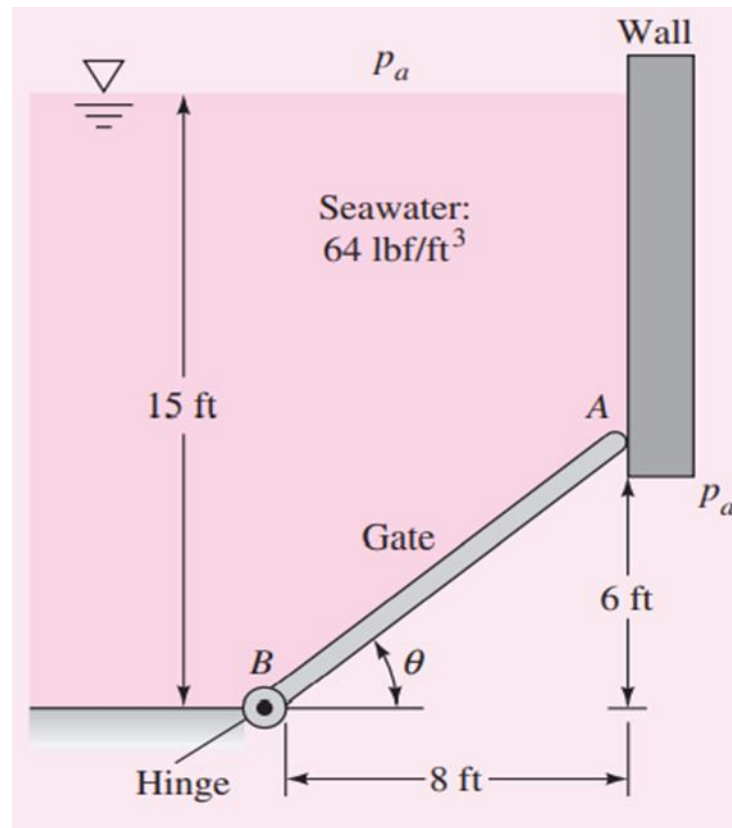
$$I_{xx} = 0.10976R^4$$

$$I_{xy} = 0$$



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- Example 5.** The gate in Fig. below is 5 ft wide, is hinged at point B, and rests against a smooth wall at point A. Compute (a) the force on the gate due to sea water pressure and center of pressure.



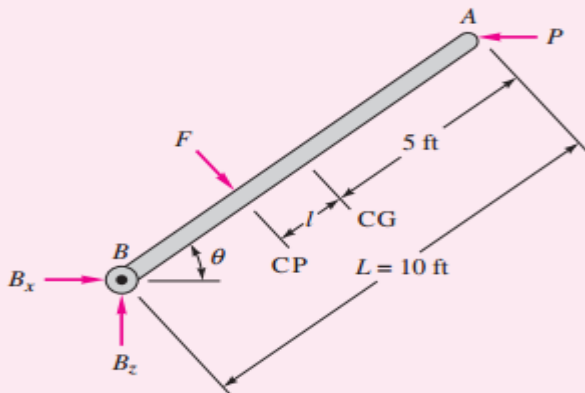


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- **Solution:**
- By geometry the gate is 10 ft long from A to B, and its centroid is halfway between, or at elevation 3 ft above point B. The depth h_{CG} is thus $15 - L - 3 = 12$ ft.

$$F = p_{CG}A = \gamma h_{CG}A = (64 \text{ lbf/ft}^3)(12 \text{ ft})(50 \text{ ft}^2) = 38,400 \text{ lbf}$$

$$I_{xy} = 0 \quad \text{and} \quad I_{xx} = \frac{bL^3}{12} = \frac{(5 \text{ ft})(10 \text{ ft})^3}{12} = 417 \text{ ft}^4 \quad l = -y_{CP} = + \frac{I_{xx} \sin \theta}{h_{CG}A} = \frac{(417 \text{ ft}^4)(\frac{6}{10})}{(12 \text{ ft})(50 \text{ ft}^2)} = 0.417 \text{ ft}$$

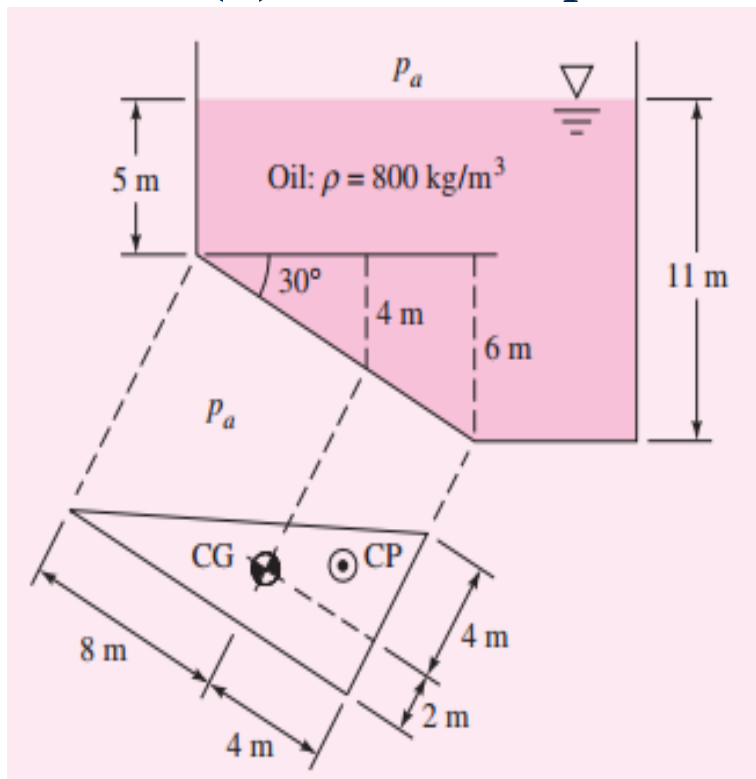


The distance from point B to force F is thus $10 - L - 5 = 4.583$ ft.



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- Example:** A tank of oil has a right-triangular panel near the bottom, as in Fig. E2.7. Omitting p_a , find the (a) hydrostatic force and (b) CP on the panel.



The centroid is one-third up (4 m) and one third over (2 m) from the lower left corner,

$A = 1/2(12)(6) = 36 \text{ m}^2$

$$I_{xx} = \frac{bL^3}{36} = \frac{(6 \text{ m})(12 \text{ m})^3}{36} = 288 \text{ m}^4$$

$$I_{xy} = \frac{b(b - 2s)L^2}{72} = \frac{(6 \text{ m})[6 \text{ m} - 2(6 \text{ m})](12 \text{ m})^2}{72} = -72 \text{ m}^4$$

$$h_{CG} = 5 + 4 = 9 \text{ m}$$

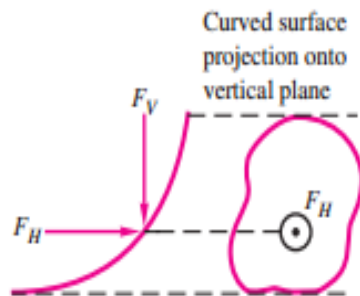
$$F = \rho g h_{CG} A = (800 \text{ kg/m}^3)(9.807 \text{ m/s}^2)(9 \text{ m})(36 \text{ m}^2) = 2.54 \times 10^6 \text{ (kg} \cdot \text{m)/s}^2 = 2.54 \times 10^6 \text{ N} = 2.54 \text{ MN}$$

$$y_{CP} = -\frac{I_{xx} \sin \theta}{h_{CG} A} = -\frac{(288 \text{ m}^4)(\sin 30^\circ)}{(9 \text{ m})(36 \text{ m}^2)} = -0.444 \text{ m}$$

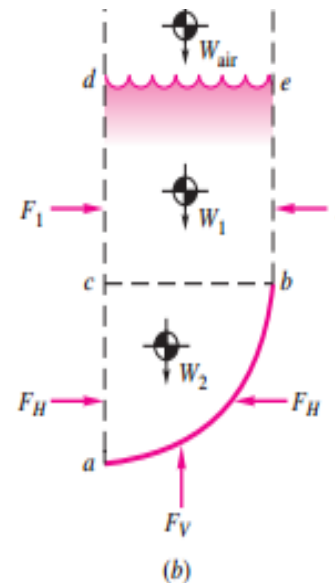
$$x_{CP} = -\frac{I_{xy} \sin \theta}{h_{CG} A} = -\frac{(-72 \text{ m}^4)(\sin 30^\circ)}{(9 \text{ m})(36 \text{ m}^2)} = +0.111 \text{ m}$$

Hydrostatic Forces on Curved Surfaces

- The horizontal component of force on a curved surface equals the force on the plane area formed by the projection of the curved surface onto a vertical plane normal to the component.
- The vertical component of pressure force on a curved surface equals in magnitude and direction the weight of the entire column of fluid, both liquid and atmosphere, above the curved surface.



(a)

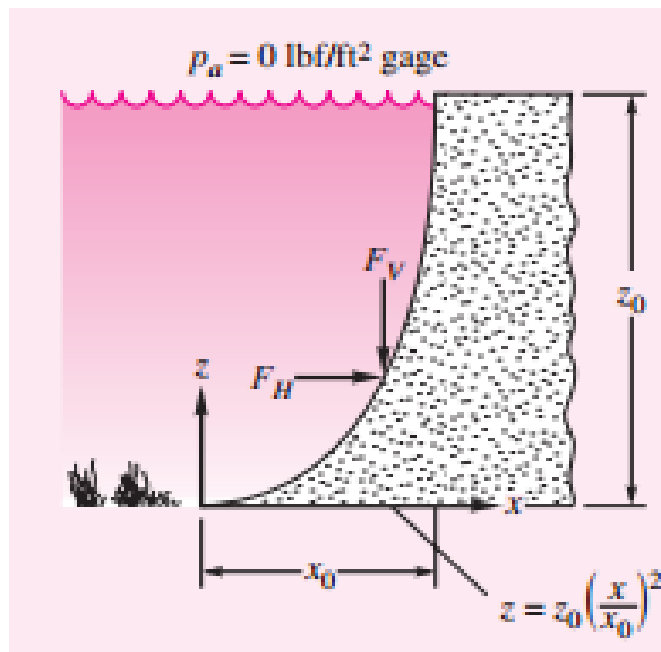


$$F_V = W_1 + W_2 + W_{air}$$



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- Example:** A dam has a parabolic shape $z/z_0 = (x/x_0)^2$ as shown in Fig. E2.8a, with $x_0 = 10$ ft and $Z_0 = 24$ ft. The fluid is water, $\gamma = 62.4$ lbf/ft³, and atmospheric pressure may be omitted. Compute the forces F_H and F_V on the dam and their line of action. The width of the dam is 50 ft.





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Solution: The vertical projection of the parabola lies along the z axis in Fig. E2.8b and is a rectangle 24 ft high and 50 ft wide. Its centroid is halfway down, or $h_{CG}=24/2=12$ ft. Its area is $A_{proj}=(24 \text{ ft})(50 \text{ ft})=1200 \text{ ft}^2$

$$F_H = \gamma h_{CG} A_{proj} = \left(62.4 \frac{\text{lbf}}{\text{ft}^3} \right) (12 \text{ ft}) (1200 \text{ ft}^2) = 898,560 \text{ lbf} \approx 899 \times 10^3 \text{ lbf}$$

The line of action of F_H is below the centroid of A_{proj} , as given by Eq. (2.29):

$$y_{CP, proj} = - \frac{I_{xx} \sin \theta}{h_{CG} A_{proj}} = - \frac{(1/12)(50 \text{ ft})(24 \text{ ft})^3 \sin 90^\circ}{(12 \text{ ft})(1200 \text{ ft}^2)} = -4 \text{ ft}$$

Thus F_H is $12 + 4 = 16$ ft, or two-thirds of the way down from the surface (8 ft up from the bottom).

Note that you calculate F_H and its line of action from the vertical projection of the parabola, not from the parabola itself. Since this projection is vertical, its angle 90



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Solution: The vertical force F_V equals the weight of water above the parabola. The weight of this parabolic amount of water is;

$$F_V = \gamma A_{\text{section}} b = \left(62.4 \frac{\text{lbf}}{\text{ft}^3} \right) \left[\frac{2}{3} (24 \text{ ft})(10 \text{ ft}) \right] (50 \text{ ft}) = 499,200 \text{ lbf} \approx 499 \times 10^3 \text{ lbf}$$

This force acts downward, through the centroid of the parabolic section, or at a distance $3x_0/8 = 3.75$ ft over from the origin. The resultant hydrostatic force on the dam is;

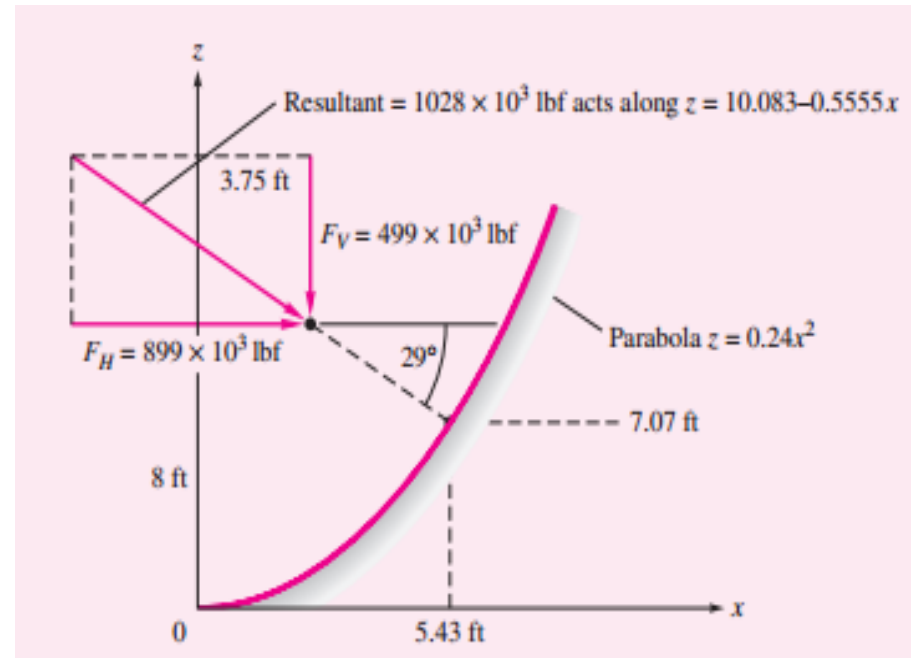
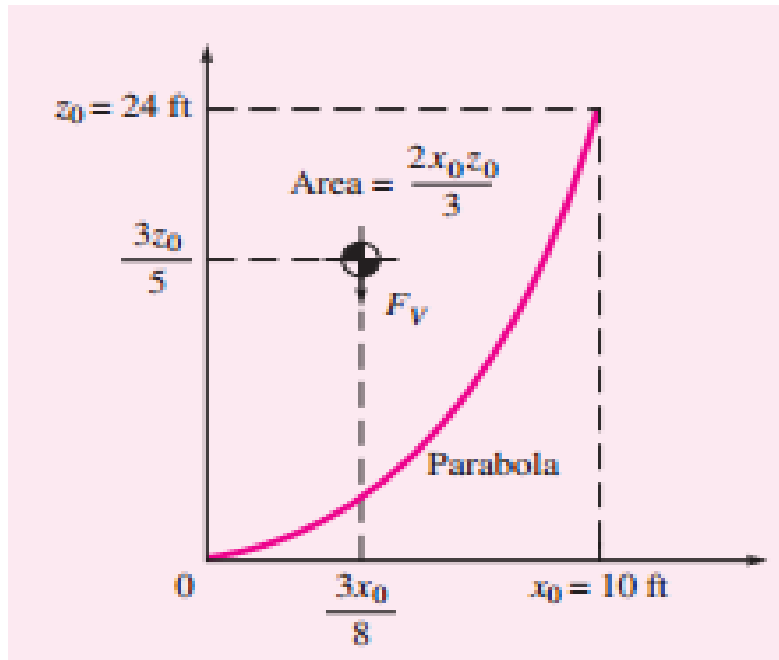
$$F = (F_H^2 + F_V^2)^{1/2} = [(899 \text{E}3 \text{ lbf})^2 + (499 \text{E}3 \text{ lbf})^2]^{1/2} = 1028 \times 10^3 \text{ lbf at } 29^\circ \text{ Ans.}$$

This resultant is shown in Fig. and passes through a point 8 ft up and 3.75 ft over from the origin. It strikes the dam at a point 5.43 ft over and 7.07 ft up, as shown.



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Solution:





Buoyancy and Stability

- The two laws of buoyancy discovered by Archimedes are;
 1. A body immersed in a fluid experiences a vertical buoyant force equal to the weight of the fluid it displaces.
 2. A floating body displaces its own weight in the fluid in which it floats.

$$F_B = \int_{\text{body}} (p_2 - p_1) dA_H = -\gamma \int (z_2 - z_1) dA_H = (\gamma)(\text{body volume})$$

= weight of fluid equivalent to body volume

The line of action of the buoyant force passes through the center of volume of the displaced body; that is, its center of mass computed as if it had uniform density. This point through which F_B acts is called the **center of buoyancy**.



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Floating bodies are a special case; only a portion of the body is submerged, with the remainder poking up out of the free surface. Not only does the buoyant force equal the body weight, but also they are collinear since there can be no net moments for static equilibrium.

$$F_B = (\gamma)(\text{displaced volume}) = \text{floating-body weight}$$

For floating bodies, the weight of the entire body must be equal to the buoyant force, which is the weight of the fluid whose volume is equal to the volume of the submerged portion of the floating body.

That is,

$$F_B = W \rightarrow \rho_f g V_{\text{sub}} = \rho_{\text{ave, body}} g V_{\text{total}} \rightarrow \frac{V_{\text{sub}}}{V_{\text{total}}} = \frac{\rho_{\text{ave, body}}}{\rho_f}$$



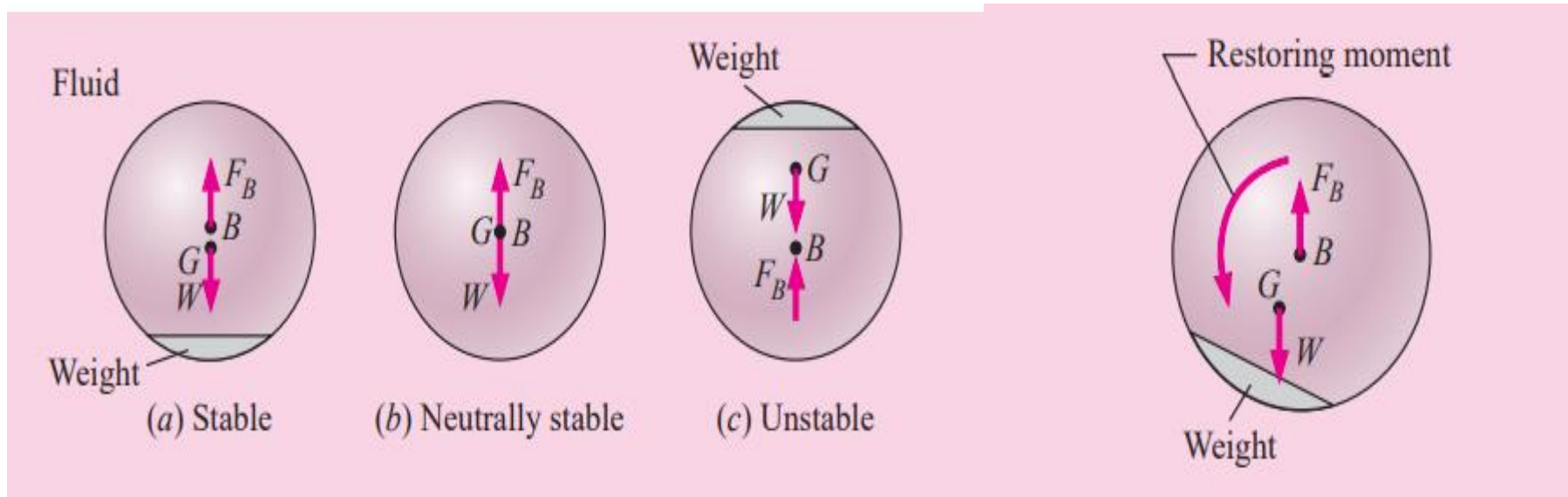
Stability of Immersed body

- The rotational stability of an immersed body depends on the relative locations of the center of gravity G of the body and the center of buoyancy B , which is the centroid of the displaced volume.
- An immersed body is **stable** if the body is bottom-heavy and thus point **G is directly below point B** .
- An immersed body whose center of gravity **G is directly above point B is unstable**, and any disturbance will cause this body to turn upside down.



...Cont'd

A body for which **G and B coincide** is **neutrally stable**. This is the case for bodies whose density is constant throughout. where the center of gravity is not vertically aligned with the center of buoyancy, It is not really appropriate to discuss stability for this case since the body is not in a state of equilibrium.



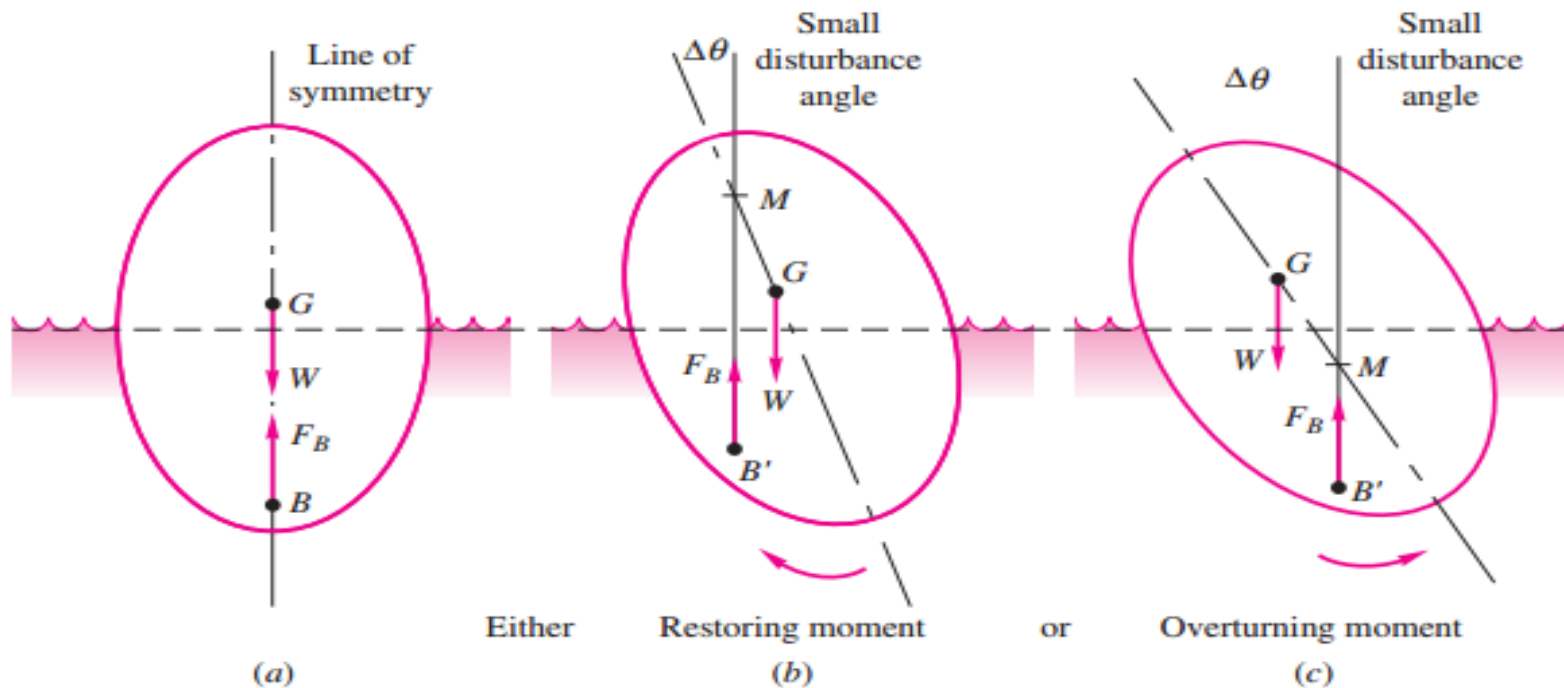


Stability of Floating body

- A measure of stability for floating bodies is **the metacentric height GM** , which is the distance between the center of gravity G and the metacenter M —the intersection point of the lines of action of the buoyant force through the body before and after rotation.
- A floating body is stable if point M is above point G , and thus GM is positive, and unstable if point M is below point G , and thus GM is negative.



...Cont'd



How to calculate the metacenter height? Do in your home



Pressure Distribution in Rigid-Body Motion

In rigid body motion:

- ❖ all particles are in combined translation and rotation,
- ❖ no relative motion between particles,
- ❖ no strains or strain rates, so that the viscous term in Eq. (2.8) vanishes,

$$\nabla p = \rho(\mathbf{g} - \mathbf{a})$$

- ❖ The pressure gradient acts in the direction $\mathbf{g}-\mathbf{a}$, and lines of constant pressure (including the free surface, if any) are perpendicular to this direction.

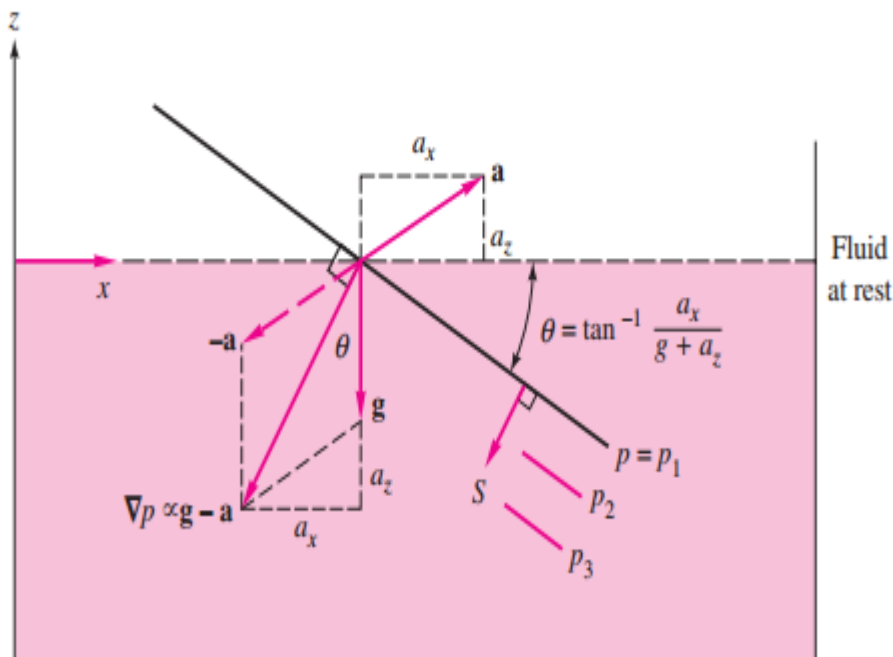


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In the case of uniform rigid-body acceleration, Eq. (2.38) applies, $\mathbf{a}$ having the same magnitude and direction for all particles.

The surfaces of constant pressure must be perpendicular to this and are thus tilted at a downward angle θ such that

$$\theta = \tan^{-1} \frac{a_x}{g + a_z}$$





...Cont'd

The rate of increase of pressure in the direction g-a is greater than in ordinary hydrostatics and is given by

$$\frac{dp}{ds} = \rho G \quad \text{where } G = [a_x^2 + (g + a_z)^2]^{1/2}$$

These results are independent of the size or shape of the container as long as the fluid is continuously connected throughout the container.



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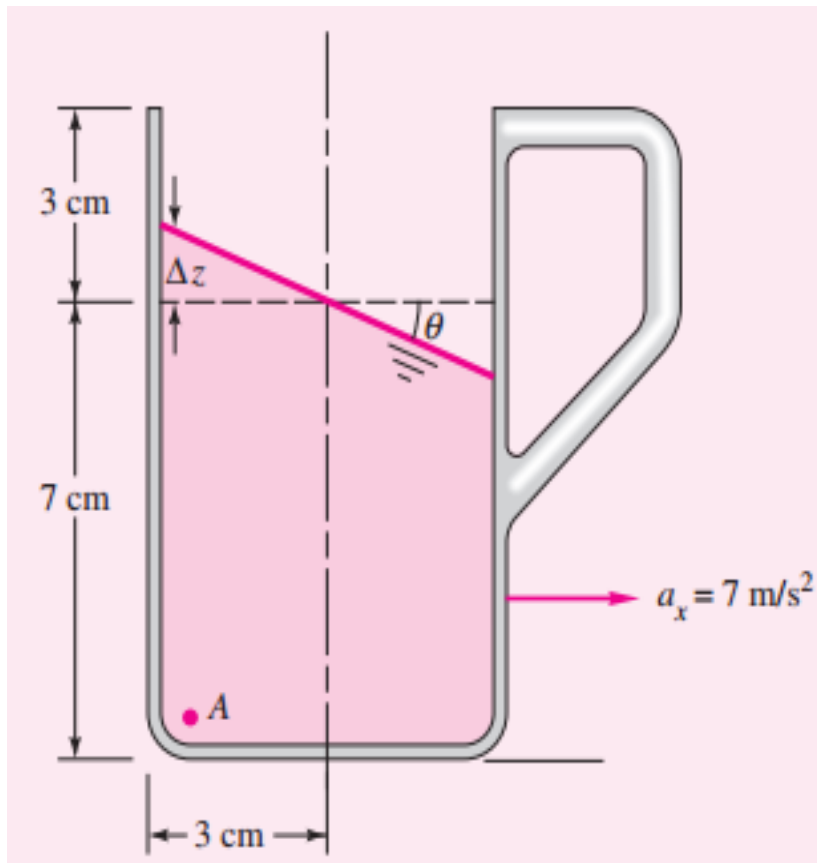
Example: A drag racer rests her coffee mug on a horizontal tray while she accelerates at 7 m/s^2 . The mug is 10 cm deep and 6 cm in diameter and contains coffee 7 cm deep at rest.

- (a) Assuming rigid-body acceleration of the coffee, determine whether it will spill out of the mug.
- (b) Calculate the gage pressure in the corner at point A if the density of coffee is 1010 kg/m^3 .



...Cont'd

Solution: Rigid-body horizontal acceleration, $a_x = 7 \text{ m/s}^2$.



$$\theta = \tan^{-1} \frac{a_x}{g} = \tan^{-1} \frac{7.0 \text{ m/s}^2}{9.81 \text{ m/s}^2} = 35.5^\circ$$

If the mug is symmetric, the tilted surface will pass through the center point of the rest position, then

$$\Delta z = (3 \text{ cm})(\tan 35.5^\circ) = 2.14 \text{ cm} < 3 \text{ cm}$$

Therefore no spilling



...Cont'd

The pressure at A can be computed from Eq. (2.40), using the perpendicular distance Δs from the surface to A.

When at rest, $p_A = \rho g h_{\text{rest}} = (1010 \text{ kg/m}^3) (9.81 \text{ m/s}^2) (0.07 \text{ m}) = 694 \text{ Pa}$.

When accelerating,

$$p_A = \rho G \Delta s = \left(1010 \frac{\text{kg}}{\text{m}^3}\right) \left[\sqrt{(9.81)^2 + (7.0)^2} \right] [(0.07 + 0.0214) \cos 35.5^\circ] \approx 906 \text{ Pa}$$

The acceleration has increased the pressure at A by 31 percent. Think about this alternative: why does it work? Since $a_z = 0$, we may proceed vertically down the left side to compute

$$p_A = \rho g (z_{\text{surf}} - z_A) = (1010 \text{ kg/m}^3) (9.81 \text{ m/s}^2) (0.0214 + 0.07 \text{ m}) = 906 \text{ Pa}$$



Rigid-Body Rotation

Consider rotation of the fluid about the z axis without any translation, as sketched in Fig. 2.22. We assume that the container has been rotating long enough at constant for the fluid to have attained rigid-body rotation. The fluid acceleration will then be a centripetal term.

the angular velocity and position vectors are given by

$$\boldsymbol{\Omega} = \mathbf{k}\Omega \quad \mathbf{r}_0 = \mathbf{i}_r r$$

Then the acceleration is given by

$$\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_0) = -r\Omega^2 \mathbf{i}_r$$

The force balance is then

$$\nabla p = \mathbf{i}_r \frac{\partial p}{\partial r} + \mathbf{k} \frac{\partial p}{\partial z} = \rho(\mathbf{g} - \mathbf{a}) = \rho(-g\mathbf{k} + r\Omega^2 \mathbf{i}_r)$$



...Cont'd

Equating like components, we find the pressure field by solving two first-order partial differential equations:

$$\frac{\partial p}{\partial r} = \rho r \Omega^2 \quad \frac{\partial p}{\partial z} = -\gamma$$

Integrate the first equation “partially,” holding z constant, with respect to r . The result is

$$p = \frac{1}{2} \rho r^2 \Omega^2 + f(z)$$

$$p = \frac{1}{2} \rho r^2 \Omega^2 + f(z)$$

Now differentiate this with respect to z and compare with the second relation

$$\frac{\partial p}{\partial z} = 0 + f'(z) = -\gamma$$

$$f(z) = -\gamma z + C$$

$$p = \text{const} - \gamma z + \frac{1}{2} \rho r^2 \Omega^2$$



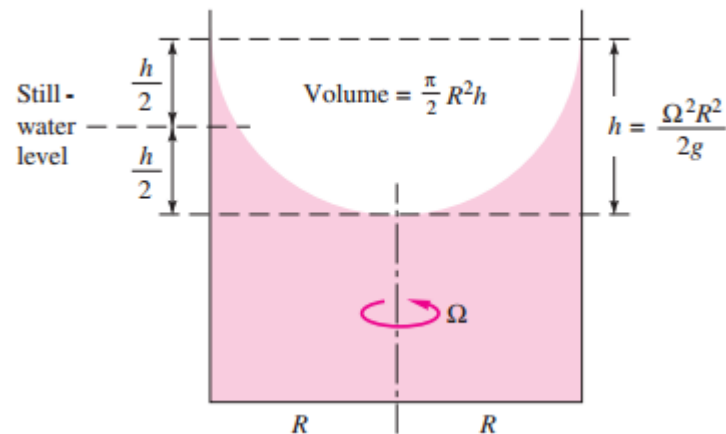
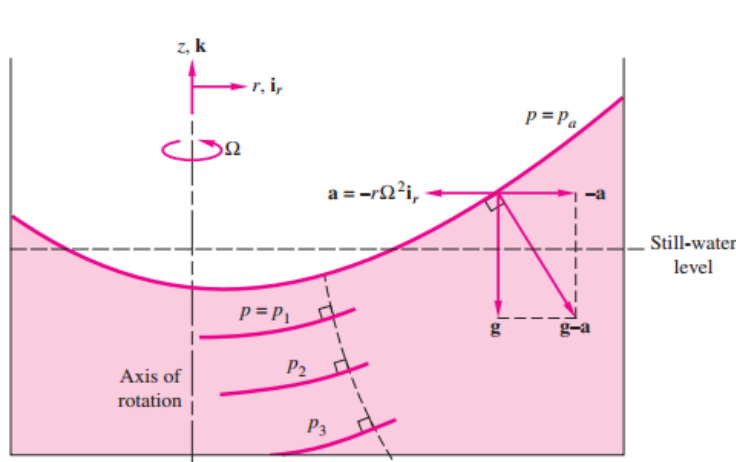
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If $p = p_0$ at $(r, z) = (0, 0)$, then $C = p_0$.

$$p = p_0 - \gamma z + \frac{1}{2}\rho r^2 \Omega^2$$

The pressure is linear in z and parabolic in r . If we wish to plot a constant-pressure surface, say, $p=p_1$

$$z = \frac{p_0 - p_1}{\gamma} + \frac{r^2 \Omega^2}{2g} = a + br^2$$





...Cont'd

Example ; The coffee cup in Example 2.13 is removed from the drag racer, placed on a turntable, and rotated about its central axis until a rigid-body mode occurs. Find (a) the angular velocity that will cause the coffee to just reach the lip of the cup and (b) the gage pressure at point A for this condition.

Solution: The cup contains 7 cm of coffee. The remaining distance of 3 cm up to the lip must equal the distance $h/2$ in Fig. 2.23. Thus

$$\frac{h}{2} = 0.03 \text{ m} = \frac{\Omega^2 R^2}{4g} = \frac{\Omega^2 (0.03 \text{ m})^2}{4(9.81 \text{ m/s}^2)}$$

Solving, we obtain

$$\Omega^2 = 1308 \quad \text{or} \quad \Omega = 36.2 \text{ rad/s} = 345 \text{ r/min}$$

To compute the pressure, it is convenient to put the origin of coordinates r and z at the bottom of the free-surface depression



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The gage pressure here is $p_0=0$, and point A is at $(r, z) = (3 \text{ cm}, 4 \text{ cm})$.

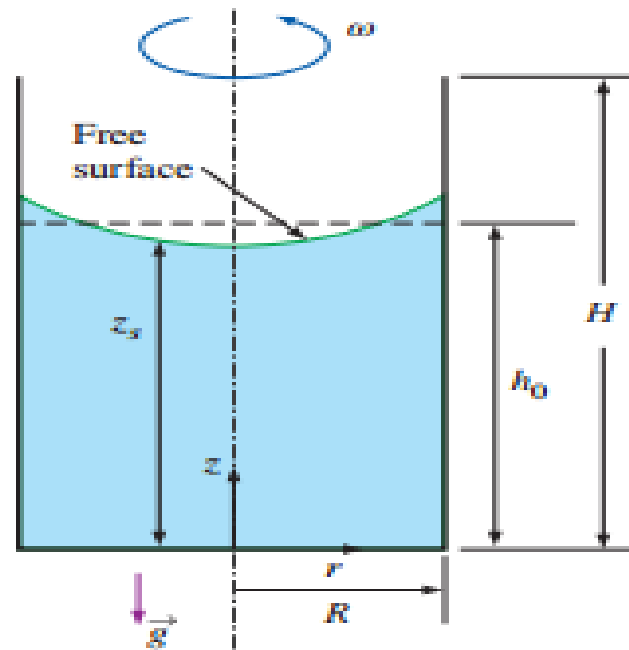
$$\begin{aligned} p_A &= 0 - (1010 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(-0.04 \text{ m}) \\ &\quad + \frac{1}{2}(1010 \text{ kg/m}^3)(0.03 \text{ m})^2(1308 \text{ rad}^2/\text{s}^2) \\ &= 396 \text{ N/m}^2 + 594 \text{ N/m}^2 = 990 \text{ Pa} \end{aligned}$$

This is about 43 percent greater than the still-water pressure $p_A = 694 \text{ Pa}$



...Cont'd

Exercise: A 20-cm-diameter, 60-cm-high vertical cylindrical container, shown in Fig. below, is partially filled with 50-cm-high liquid whose density is 850 kg/m^3 . Now the cylinder is rotated at a constant speed. Determine the rotational speed at which the liquid will start spilling from the edges of the container.





Pressure Measurement

Instruments may be grouped into four categories:

1. Gravity-based: barometer, manometer, deadweight piston.
2. Elastic deformation: bourdon tube (metal and quartz), diaphragm, bellows, strain-gage, optical beam displacement.
3. Gas behavior: gas compression (McLeod gage), thermal conductance (Pirani gage), molecular impact (Knudsen gage), ionization, thermal conductivity, air piston.



...Cont'd

4. Electric output: resistance (Bridgman wire gage), diffused strain gage, capacitative, piezoelectric, potentiometric, magnetic inductance, magnetic reluctance, linear variable differential transformer (LVDT), resonant frequency.
5. Luminescent coatings for surface pressure

Please read each category in detail



Thank You
End of Chapter Two